NKUTHALO MAKONCO

Electrical & Automation Technician | PLC & Control Systems Specialist

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PROFESSIONAL SUMMARY

Dedicated and results-driven Electrical & Automation Technician with 9 years of progressive, hands-on experience in industrial PLC programming, SCADA/HMI development, robotics integration, and electrical systems maintenance. Proven track record of minimising downtime, improving system reliability, and delivering automation solutions across manufacturing environments. Adept at programming Siemens, Allen-Bradley, and Schneider Electric platforms and proficient in KUKA robot systems. Strong foundation in project management, technical documentation, and cross-functional collaboration.

PROFESSIONAL EXPERIENCE

Electrical & Robot Technician — First Battery

September 2023 – Present (2 yrs 7 mos) • East London, South Africa | On-site

- Program and configure Siemens, Allen-Bradley, and Schneider Electric PLCs to automate and optimise manufacturing processes.
- Program and maintain KUKA robot systems for automated material handling and production line operations.
- Develop and maintain SCADA/HMI applications to enable real-time process monitoring and operator control.
- Manage industrial network topologies and troubleshoot PLC/HMI faults to minimise system downtime.
- Provide technical support for all automation systems and conduct informal training for operators and maintenance staff.
- Agree project plans, budgets, and timelines with management; oversee execution of improvement and upgrade projects.
- Apply Material Requirements Planning (MRP) to align maintenance activities with production schedules.

Electrical Engineering Technician — First Battery

March 2019 – September 2023 (4 yrs 7 mos) • East London, South Africa

- Designed, installed, commissioned, and maintained electrical control systems, instrumentation, and automation equipment.
- Performed root-cause fault diagnosis on PLCs, HMIs, and electrical systems; implemented corrective and preventative maintenance.
- Led and participated in annual shutdown maintenance programmes according to scheduled plans.
- Conducted machine improvement and upgrade projects, including retrofitting of electrical equipment and control systems.
- Performed routine safety inspections on electrical equipment to ensure compliance with operational and safety standards.
- Interpreted and produced technical drawings, schematics, and CAD documentation for electrical systems.

Assistant Electrician — First Battery

March 2016 – February 2019 (3 yrs) • East London, South Africa

- Assisted qualified artisans with installation, maintenance, and repair of electrical machinery and equipment.
- Responded to breakdown and work order requests, carrying out timeous repairs to reduce operational downtime.
- Conducted routine inspections on electrical equipment to ensure reliability and safety adherence.
- Supported preventative and corrective maintenance programmes across various production machinery.

EDUCATION

Bachelor of Technology — Electrical Engineering
University of South Africa (UNISA) | 2018

National Diploma — Electrical Engineering
Cape Peninsula University of Technology (CPUT) | 2016

Bachelor of Science — Computing
University of South Africa (UNISA) | 2025

TECHNICAL SKILLS

PLC Platforms: Siemens Simatic S7 / TIA Portal, Allen-Bradley / Rockwell, Schneider Electric
Robotics: KUKA Robot Programming & Maintenance
SCADA / HMI: Design, Development & Commissioning
Electrical Systems: Control systems, instrumentation, automation technologies, power distribution
CAD Tools: Technical drawing, schematic interpretation and production
Programming Languages: C++, Python, Microsoft Visual Basic, SQL
Software: Microsoft Office Suite, MRP Systems
Networking: Industrial network topologies, Industrial PCs, Fieldbus communication

CORE COMPETENCIES

PLC Programming & Automation • System Integration & Commissioning • SCADA/HMI Development
Fault Diagnosis & Troubleshooting • Preventative & Corrective Maintenance • Root Cause Analysis
Project Management • Technical Documentation • Operator Training & Support • Safety Compliance

References available on request